

Cálculo II Lista 3 - Jeann Victor Batista

1) Calcule a integral fazendo a substituição $u = g(x)$ e $du = g'(x)dx$

a) $\int 2(2x+4)^5, u = 2x+4.$

$$u-4 = 2x \rightarrow u-4/2 = x.$$

$$du = 2 \cdot dx \rightarrow dx = du/2.$$

Substituição:

$$\int 2 \cdot u^5 \cdot \frac{du}{2} \rightarrow \int u^5 \rightarrow \frac{u^6}{6}.$$

$$\frac{u^6}{6} = \frac{(2x+4)^6}{6} \text{ é a primitiva.}$$

b) $\int 7\sqrt{7x-1}dx, u = 7x-1$

$$u+1 = 7x \rightarrow x = (u+1)/7$$

$$du = 7dx \rightarrow dx = du/7.$$

Substituição

$$\int 7 \cdot (u)^{1/2} \cdot \frac{du}{7} \rightarrow \int u^{1/2} \cdot du \rightarrow \int u^{1/2} \cdot du$$

$$\frac{u}{\frac{3}{2}} \rightarrow \frac{2u}{3} \rightarrow \frac{2}{3} \cdot u^{3/2} = \frac{2}{3} \cdot (7x-1)^{3/2}$$

$$2/3 \cdot (7x-1)^{3/2} \text{ é a primitiva.}$$

c) $\int \frac{dt}{(1-6t)^4}, u = 1-6t$

$$u = 1-6t \rightarrow u-1 = -6t \rightarrow (u-1)/-6 = t$$

$$du = -6dt \rightarrow \frac{-du}{6} = dt.$$

Substituição:

$$\int \frac{-du/6}{(u)^4} \rightarrow \int \frac{-du}{6u^4} \rightarrow \int \frac{-du}{6u^4} \rightarrow \int \frac{-du}{6u^4}$$

$$-\frac{1}{6} \cdot \int u^{-4} \cdot du \rightarrow -\frac{1}{6} \cdot \frac{u^{-3}}{-3} \rightarrow \frac{+u^{-3}}{18} \rightarrow \frac{1}{18u^3}$$

$$\frac{1}{18(1-6t)^3} \rightarrow \text{Primitiva.}$$

$$d) \int \cos 3x \, dx, u=3x$$

$$u=3x, \quad u/3 = x.$$

$$du = 3 \, dx = (du)/3 = dx.$$

Substituição

$$\int \cos u \cdot \frac{1}{3} \cdot du \rightarrow \frac{1}{3} \int \cos u \cdot du \rightarrow \frac{1}{3} \sin u + C$$

$$\frac{1}{3} \sin 3x + C \rightarrow \text{Primitiva}$$

$$e) \int 2x(x^2+5)^{-3} \, dx, u=x^2+5$$

$$u = x^2 + 5 \rightarrow u-5 = x^2 \rightarrow \sqrt{u-5} = x$$

$$du = 2x \, dx \rightarrow$$

Substituição

$$\int 2x(x^2+5)^{-3} \, dx \rightarrow \int u^{-3} \cdot du \rightarrow \left(\frac{u^{-3}}{-3} \right) \rightarrow \frac{1}{-3u^3} \rightarrow -\frac{1}{3} \cdot \frac{1}{u^3}$$

$$-\left(\frac{1}{3 \cdot (x^2+5)^3} \right) \rightarrow -\frac{1}{3 \cdot (x^2+5)^3} = \text{Primitiva}$$

$$f) \int \frac{4x^3}{(x^4+1)^2} \, dx, u=x^4+1$$

$$u = x^4 + 1 \rightarrow u-1 = x^4 \rightarrow \sqrt[4]{u-1} = x$$

$$du = 4x^3 \, dx$$

Substituição

$$\int \frac{du}{u^2} \rightarrow \int u^{-2} \cdot du \rightarrow -\frac{u^{-1}}{1} \rightarrow -(x^4+1)^{-1} + C$$

$$-\frac{1}{(x^4+1)} + C$$

$$g) \int \frac{(1+\sqrt{x})^{1/3}}{\sqrt{x}} dx, u=1+\sqrt{x}$$

$$u=1+\sqrt{x} \rightarrow u-1=\sqrt{x} \rightarrow \frac{1}{2} \cdot x^{-1/2}$$

$$du = x^{-1/2} \rightarrow du = x^{-1/2} \rightarrow \frac{1}{2\sqrt{x}} dx \rightarrow -2\sqrt{x} du \rightarrow 2\sqrt{x} du.$$

Substituição.

$$\int (u)^{1/3} \cdot 2\sqrt{x} du \rightarrow 2 \cdot \int u^{1/3} du \rightarrow 2 \cdot \frac{u^{4/3}}{4/3} \rightarrow \frac{3}{2} u^{4/3}$$

$$\text{Logo } \frac{3}{2} (1+\sqrt{x})^{4/3} + C \text{ é a primitiva}$$

$$h) \int x \cdot \sin(ax^2) dx, u=ax^2$$

$$du = 2x dx \rightarrow dx = du/2x$$

Substituição

$$\int x \cdot \sin(u) \cdot \frac{1}{2} \frac{du}{x} \rightarrow \frac{1}{2} \int \sin(u) \cdot du \rightarrow -\frac{1}{2} \cos(u) du$$

$$-\frac{1}{2} \cos(ax^2) + C \rightarrow \text{Primitiva}$$

$$i) \int \sec(2t) \tan(2t) dt, u=2t$$

$$du = 2 dt \rightarrow dt = du/2$$

Substituição

$$\int \sec(u) \cdot \tan(u) \cdot \frac{du}{2} \rightarrow \frac{1}{2} \cdot \int \sec u \cdot \tan(u) \cdot du = \sec(u)$$

$$\frac{1}{2} \cdot \sec(u) + C$$

$$\frac{1}{2} \cdot \sec(2t) + C \rightarrow \text{Primitiva}$$

$$j) \int \frac{9x^2 dx}{\sqrt{1-x^3}}, u=1-x^3$$

$$9x^2 \cdot dx \cdot (1-x^3)^{-1/2}$$

$$du = -3x^2 dx \rightarrow -\frac{1}{3} \cdot du = x^2 dx \rightarrow \text{Substituição}$$

$$\int \frac{9 \cdot x^2 dx}{\sqrt{u}} \rightarrow \int \frac{9}{\sqrt{u}} \cdot \frac{1}{3} du \rightarrow \int \frac{3}{\sqrt{u}} \cdot du \rightarrow 3 \cdot \int u^{-1/2}$$

$$-3 \cdot \frac{u^{1/2}}{1/2} \rightarrow -3 \cdot 2 \cdot u^{1/2} \rightarrow -6 u^{1/2} \rightarrow -6 (1-x^3)^{1/2} + C$$

$$1 - x^4 = -\frac{2}{3}x^3$$

K) $\int \sqrt{x} \cdot \sin(x^{3/2} - 1) dx$, $u = x^{3/2} - 1$.

$$du = \frac{3}{2} x^{1/2} dx \rightarrow \frac{2du}{3} = x^{1/2} dx$$

Substituição:

$$\int x^{1/2} \cdot \sin(u) \cdot dx \rightarrow \int \sin(u) \cdot x^{1/2} \cdot dx \rightarrow \int \sin(u) \cdot \frac{2}{3} du$$

$$\frac{2}{3} \cdot \int \sin u \, du \rightarrow \frac{2}{3} \cdot (-\cos(u)) \rightarrow -\frac{2}{3} \cos(u) + C$$

$$\text{Primitiva} = -\frac{2}{3} \cdot (\cos(x^{3/2} - 1) + C)$$

L) $\int \cos^3 \theta \cdot \sin \theta \, d\theta$, $u = \cos \theta$

$$du = -\sin \theta \, d\theta \rightarrow -du = \sin \theta \, d\theta$$

$$\int u^3 \cdot (-du) \rightarrow -\int u^3 \rightarrow -\frac{u^4}{4} \rightarrow -\frac{\cos^4 \theta}{4}$$

M) $\int \frac{\sec^2(1/x)}{x^2} dx$, $u = 1/x$

$$u = x^{-1} \rightarrow du = -x^{-2} dx \rightarrow -du = \frac{1}{x^2} dx$$

$$\int \frac{\sec^2(u)}{x^2} dx \rightarrow \int \sec^2(u) \cdot (-du) \rightarrow -\int \sec^2(u) \cdot du$$

$$-\tan(u) + C \rightarrow -\tan(1/x) + C \rightarrow \text{Primitiva}$$

② Determine as integrais usando a regra da substituição

a) $\int x^2 \sin x^3 dx$, $u = x^3$

$$du = x^2 dx \rightarrow \int \sin u \cdot du \rightarrow -\cos u + C$$

Substituição

$$\int \sin u \cdot \frac{1}{3} du \rightarrow \frac{1}{3} \cdot \int \sin u \cdot du \rightarrow \frac{1}{3} \cdot (-\cos u) + C$$

$$-\frac{1}{3} \cdot \cos(x^3) + C \rightarrow \text{Primitiva}$$

b) $x^3 \sqrt{1-x^4} dx$. $u = (1-x^4)$

$du = (1-x^4)' \rightarrow du = -4x^3 dx \rightarrow \frac{du}{-4} = x^3 dx$

Substituição:

$\int (u)^{1/2} \cdot x^3 dx \rightarrow \int (u)^{1/2} \cdot \frac{1}{-4} du \rightarrow -\frac{1}{4} \int u^{1/2} du$

$-\frac{1}{4} \cdot \frac{u^{3/2}}{3/2} \rightarrow -\frac{1}{4} \cdot \frac{2}{3} u^{3/2} \rightarrow -\frac{1}{6} u^{3/2}$

$-\frac{1}{6} (1-x^4)^{3/2} + C \rightarrow \text{Primitiva}$

c) $\int \cos \sqrt{x} dx \rightarrow u = (\sqrt{x})$

$du = \frac{1}{2\sqrt{x}} dx \rightarrow du = \frac{1}{2} x^{-1/2} dx \rightarrow 2 du = \frac{1}{\sqrt{x}} dx \rightarrow 2\sqrt{x} du = dx$

Substituição

$\frac{\cos u}{\sqrt{x}} \cdot 2\sqrt{x} \rightarrow 2 \cdot \frac{\cos u}{\sqrt{x}} \cdot \sqrt{x} \rightarrow 2 \cdot \cos u \rightarrow 2 \cdot \sin(\sqrt{x}) + C$

A primitiva é: $2 \cdot \sin(\sqrt{x}) + C$.

$S_0' \rightarrow \left(\frac{-1}{6} \right) \cdot (1-1^4)^{3/2} - \left(\frac{-1}{6} \right) \cdot (1-0^4)^{3/2}$

$0 - \left(\frac{-1}{6} \right) \cdot (1)^{3/2} \rightarrow 0 + \frac{1}{6} = \frac{1}{6}$

d) $\int \sqrt[4]{3x-1} dx$ $u = 3x-1$

$du = (3x-1)' \rightarrow 3 dx \rightarrow du/3 = dx$

Substituição

$\int (u)^{1/3} \cdot du \cdot \frac{1}{3} \rightarrow \frac{1}{3} \cdot \frac{u^{4/3}}{4/3} \rightarrow \frac{1}{3} \cdot \frac{3}{4} u^{4/3} \rightarrow \frac{u^{4/3}}{4} + C$

$\frac{(3x-1)^{4/3}}{4} + C \rightarrow \text{Primitiva}$

$$e) \int \cos(5x+2) dx \quad u = (5x+2)$$

$$du = (5x+2)' \rightarrow 5 dx \rightarrow du/5 = dx$$

Substituição

$$\int \cos(u) \cdot du \cdot \frac{1}{5} \rightarrow \frac{1}{5} \cdot \sin(u) \cdot du \rightarrow \frac{1}{5} \cdot \sin(u) + C$$

$$= \frac{1}{5} \cdot \sin(5x+2) + C \rightarrow \text{Primitiva}$$

$$f) \int x / \sqrt{x^2+4} dx \quad u = x^2+4$$

$$du = (x^2+4)' \rightarrow 2x dx \rightarrow du/2 = x dx$$

Substituição

$$\int \frac{x}{u^{1/2}} dx \rightarrow \int u^{-1/2} \cdot x dx \rightarrow \int u^{-1/2} \cdot \frac{1}{2} du \rightarrow \frac{1}{2} \int u^{-1/2} du$$

$$\frac{1}{2} \cdot \frac{u^{1/2}}{1/2} \rightarrow \frac{1}{2} \cdot 2 u^{1/2} \rightarrow u^{1/2} \rightarrow (x^2+4)^{1/2} + C$$

$$\text{Primitiva} = (x^2+4)^{1/2} + C$$

$$g) \int \ln x / x dx \quad u = \ln x$$

$$du = (\ln x)' \rightarrow du = 1/x dx$$

Substituição

$$\int u \cdot \frac{1}{x} dx \rightarrow \int u \cdot du \rightarrow \frac{u^2}{2} + C$$

$$\text{Primitiva} \rightarrow \frac{(\ln x)^2}{2} + C$$

$$h) \int \cot x dx \rightarrow \int \frac{\cos x}{\sin x} dx \quad u = \sin x$$

$$du = (\sin x)' = \cos x dx \rightarrow du = \cos x dx$$

Substituição

$$\int \frac{\cos x}{u} dx \rightarrow \int \frac{1}{u} \cdot \cos x dx \rightarrow \int \frac{1}{u} du \rightarrow \int u^{-1} du$$

$$\frac{u^{-1/2}}{-1/2} \rightarrow -2 u^{-1/2} \rightarrow -2 (\sin x)^{-1/2} + C \rightarrow \text{Primitiva}$$

$$2) \int_0^2 (x^2/x^3+1) dx \rightarrow u = (x^3+1)$$

$$du = (x^3+1)' \rightarrow du = 3x^2 dx \rightarrow du/3 = x^2 dx$$

Substituição:

$$\int_0^2 \frac{1}{u} \cdot x^2 dx \rightarrow \int_0^2 \frac{1}{u} \cdot \frac{1}{3} du \rightarrow \frac{1}{3} \int_0^2 \frac{1}{u} du$$

$$\frac{1}{3} \ln|u| \rightarrow \frac{1}{3} \ln(x^3+1) + C \text{ primitiva sem módulo por conta intervalo}$$

$$1/3 \cdot \ln(2^3+1) - 1/3 \cdot \ln(0^3+1) \rightarrow 1/3 \cdot \ln(9) - 1/3 \cdot 0$$

$$1/3 \cdot \ln(9) \rightarrow \text{Resultado}$$

$$3) \int_0^2 \frac{3 dx}{x \ln^2 3x}$$

$$\text{Reescrevendo a expressão} \rightarrow 3 \int_0^2 (1/x \cdot (\ln 3x)^{-2}) dx$$

$$u = (\ln 3x) \rightarrow du = (\ln 3x)' = 1/x dx \rightarrow du$$

Substituição

$$3 \int \frac{1}{u^2} \cdot \frac{1}{x} dx \rightarrow 3 \int \frac{1}{u^2} du \rightarrow 3 \int u^{-2} du \rightarrow -3 \cdot u^{-1} + C$$

$$\text{Primitiva} = -3(\ln 3x)^{-1} + C$$

Intervalo de 1 a 2.

$$-3 \cdot \ln 3 \cdot 2 - (-3) \cdot \ln 3 \cdot 1$$

$$-3 \cdot (\ln 6) + 3 \cdot (\ln 3)$$

$$3 \cdot \left(\frac{(-1)}{\ln 6} + \frac{(1)}{\ln 3} \right)$$

$$K) \int_1^2 x e^{3x^3} dx$$

$$u = 3x^3$$

$$du = (3x^2) \rightarrow 6x \cdot dx \rightarrow du/6 = x dx$$

Substituição

$$\int_1^2 e^u \cdot x dx \rightarrow \int_1^2 e^u \cdot du \cdot \frac{1}{6} \rightarrow \frac{1}{6} \int_1^2 e^u \cdot du \rightarrow \frac{1}{6} e^u + \frac{1}{6} e^{(3x^3)}$$

Intervalo de 1 a 2.

$$\frac{1}{6} \cdot e^{3 \cdot 2^3} - \frac{1}{6} \cdot e^{3 \cdot 1^3} \rightarrow \frac{1}{6} \cdot e^{12} - \frac{1}{6} \cdot e^3 \rightarrow \frac{1}{6} (e^{12} - e^3)$$

$$L) \int_0^3 2x 3^{x^2} dx$$

$$u = x^2$$

$$du = 2x dx \rightarrow$$

Substituição:

$$\int_0^3 3^u \cdot 2x dx \rightarrow \int_0^3 3^u \cdot du \rightarrow 3^u / \ln(3)$$

$$3^{x^2} \cdot \ln(3) + C \rightarrow \text{Primitiva}$$

Intervalo de 0 a 3

$$\frac{3^9}{\ln 3} - \frac{3^0}{\ln 3} \rightarrow \frac{3^9 - 1}{\ln 3}$$

$$m) \int \frac{1}{x^2} \sqrt{2-x} dx \rightarrow \int x^{-2} \cdot (2-x)^{1/2} dx \quad u = 2-x^{-1}$$

$$du = -x^{-2} \rightarrow x^{-2} dx$$

Substituição:

$$\int u^{1/2} \cdot x^{-2} dx \rightarrow \int u^{1/2} \cdot du \rightarrow \frac{u^{1/2+1/2}}{\frac{1}{2}+1/2} \rightarrow \frac{2u^{1/2}}{1}$$

$$2/3 (2-x)^{3/2} \rightarrow \text{primitiva}$$

$$n) \int \sec x \, dx$$

$$\frac{\sec x (\sec x + \tan x)}{(\sec x + \tan x)} \rightarrow u = \sec x + \tan x \rightarrow du = (\sec x \tan x + \sec^2 x) dx$$

$$\int \frac{\sec^2 x + \sec x \tan x}{(\sec x + \tan x)} dx \rightarrow \int \frac{du}{u} \rightarrow \ln|u| + C$$

$$\text{Primitiva: } (\ln|\sec x + \tan x| + C)$$

$$o) \int \frac{1}{\sqrt{t}} \cos(\sqrt{t} + 3) dt \quad u = (\sqrt{t} + 3)$$

$$du = \frac{1}{2\sqrt{t}} dt \rightarrow \frac{1}{2} du = \frac{1}{\sqrt{t}} dt \rightarrow du = \frac{2}{\sqrt{t}} dt$$

Substituição:

$$\int \cos(u) \cdot \frac{1}{2} du \rightarrow \int \cos u \, du \rightarrow -\sin u + C$$

$$-\sin(\sqrt{t} + 3) + C \rightarrow \text{Primitiva.}$$

$$p) \int_1^4 \frac{(1 + \sqrt{x})^{11/2}}{\sqrt{x}} dx \quad u = (1 + \sqrt{x})$$

$$du = \frac{1}{2\sqrt{x}} dx \rightarrow \frac{1}{2} du = \frac{1}{\sqrt{x}} dx \rightarrow 2 du = \frac{1}{\sqrt{x}} dx$$

Substituição:

$$\int_1^4 (u)^{11/2} \cdot \frac{1}{\sqrt{x}} dx \rightarrow \int_1^4 (u)^{11/2} \cdot 2 \, du \rightarrow 2 \int_1^4 (u)^{11/2} \, du$$

$$2 \cdot \frac{u^{13/2}}{13/2} \rightarrow \frac{4}{13} u^{13/2} \rightarrow \frac{4}{13} (1 + \sqrt{x})^{13/2} + C \rightarrow \text{Primitiva}$$

Intervalo de t e x .

$$\frac{4}{13} (1 + \sqrt{4})^{13/2} - \frac{4}{13} (1 + \sqrt{1})^{13/2}$$

$$\frac{4}{13} (3)^{13/2} - \frac{4}{13} (2)^{13/2} \rightarrow \frac{4}{13} (\sqrt{27}) - \frac{4}{13} (\sqrt{8})$$

$$\frac{4}{13} (\sqrt{27} - \sqrt{8})$$

$$q) \int_0^{\pi/2} 3 \sin x \cos x \sqrt{1+3\sin^2 x} dx \quad (v = (1+3\sin^2 x))$$

$$du = (1+3\sin^2 x) \rightarrow 6 \sin x \cos x dx \quad du/6 = \sin x \cos x$$

Substituição

$$\int_0^{\pi/2} 3 \sin x \cos x \sqrt{u} dx \rightarrow 3 \int_0^{\pi/2} u^{1/2} \cdot \sin x \cos x dx$$

$$3 \int_0^{\pi/2} u^{1/2} \cdot du \cdot \frac{1}{6} \rightarrow \frac{3}{6} \cdot \int_0^{\pi/2} u^{1/2} \cdot du$$

$$\frac{3}{6} \cdot \frac{u^{3/2}}{\frac{3}{2}} \rightarrow \frac{3}{6} \cdot \frac{2u^{3/2}}{3} \rightarrow \frac{1}{3} u^{3/2} \rightarrow \frac{1}{3} (1+3\sin^2 x)^{3/2} + C$$

Intervalo de 0 a $\pi/2$.

$$\frac{1}{3} \cdot (1+3\sin^2(\pi/2))^{3/2} - \frac{1}{3} \cdot (1+3\sin^2(0))^{3/2}$$

$$\frac{1}{3} \cdot (1+3)^{3/2} - \frac{1}{3} \cdot (1+3 \cdot 0)^{3/2} \rightarrow \frac{1}{3} \cdot (4)^{3/2} - \frac{1}{3} \cdot 1$$

$$\frac{1}{3} \cdot \sqrt{64} - \frac{1}{3} \rightarrow \frac{1}{3} \cdot 8 - \frac{1}{3} \rightarrow \frac{8}{3} - \frac{1}{3} = \frac{7}{3}$$

$$r) \int_0^{\pi/2} e^{\sin^2 \theta} \cdot \sin 2\theta d\theta \quad u = \sin^2 \theta$$

$$du = \sin 2\theta \rightarrow du = \sin 2(\theta) d\theta$$

Substituição

$$\int_0^{\pi/2} e^u \cdot du \rightarrow \int e^u + C \rightarrow e^{\sin^2 \theta}$$

Intervalo de 0 a $\pi/2$.

$$e^{\sin^2(\pi/2)} - e^{\sin^2(0)} \rightarrow e^1 - e^0 \rightarrow (e-1)$$

Portanto, o resultado é: $(e-1)$.

3. Calcule a área assinalada nas figuras abaixo.

a) Intervalo de $(-2, 0)$

$$\int x \sqrt{4-x^2} \rightarrow u = 4-x^2 \quad du = -2x dx \rightarrow -\frac{du}{2} = x dx$$

Substituição:

$$\int_{-2}^0 \sqrt{u} \cdot x dx \rightarrow \int_{-2}^0 u^{1/2} \cdot \left(-\frac{1}{2}\right) du \rightarrow -\frac{1}{2} \int_{-2}^0 u^{1/2} du$$

$$= -\frac{1}{2} \cdot \frac{u^{3/2}}{3/2} \rightarrow -\frac{1}{2} \cdot \frac{2u^{3/2}}{3} \rightarrow -\frac{2}{6} u^{3/2} \rightarrow -\frac{1}{3} (4-x^2)^{3/2}$$

Intervalo de $(-2, 0)$

$$-\frac{1}{3} (4 - (-2)^2)^{3/2} - \left(-\frac{1}{3}\right) (4 - 0^2)^{3/2}$$

$$-\frac{1}{3} (4 - 4)^{3/2} - \left(-\frac{1}{3}\right) (4)^{3/2}$$

$$-\frac{1}{3} \cdot 0 - \left(-\frac{1}{3}\right) \cdot \sqrt{64} \rightarrow +\frac{1}{3} \cdot 8 \rightarrow +\frac{8}{3}$$

Intervalo de $(0, 2)$

$$-\frac{1}{3} (4 - (2)^2)^{3/2} - \left(-\frac{1}{3}\right) (4 - 0^2)^{3/2}$$

$$-\frac{1}{3} \cdot 0^{3/2} + \frac{1}{3} (4)^{3/2} \rightarrow \frac{1}{3} \cdot \sqrt{64} \rightarrow \frac{1}{3} \cdot 8 = \frac{8}{3}$$

Portanto a área tem o valor é de $\frac{8}{3} + \frac{8}{3} = \frac{16}{3}$.

b) Intervalo de 0 a π

$$y = (1 - \cos x) \text{ em } x \rightarrow u = (1 - \cos x), du = \sin x dx$$

$$\int u \cdot du \rightarrow \frac{u^2}{2} \rightarrow \frac{1}{2} (1 - \cos x)^2$$

π

$$\frac{1}{2} (1 - \cos \pi)^2 - \frac{1}{2} (1 - \cos 0)^2 \rightarrow \frac{1}{2} (1 - (-1))^2 - \frac{1}{2} (1 - 1)^2$$

$$\frac{1}{2} \cdot 2^2 - \frac{1}{2} \cdot 0^2 \rightarrow \frac{4}{2} - 0 \rightarrow 2$$

2) Calcule a integral usando a integração por partes com escolhas de u e dv indicadas que $\int u dv = uv - \int v du$

a) $\int \ln x dx$

$u = \ln x$, $du = 1/x dx$, $dv = dx$, $v = x$

Aplicando na fórmula:

$\int u dv = \ln(x) \cdot x - \int x \cdot 1/x dx$

$\int u dv = x \cdot \ln(x) - \int x/x dx \rightarrow x \cdot \ln(x) - \int 1 dx$

Primitiva $x \cdot \ln(x) - x + C$.

b) $\int x \cdot \operatorname{sen} x dx$

$u = x$, $du = 1 dx$, $dv = \operatorname{sen} x dx$, $v = -\cos x$

$\int u dv = -x \cdot \cos x + \int \cos x \cdot 1 dx$

$\int u dv = x \cdot \cos x + (1 \cdot \operatorname{sen} x) \rightarrow -x \cdot \cos x + \operatorname{sen} x + C$

A primitiva é: $-x \cdot \cos x + \operatorname{sen} x + C$

c) $\int t^3 \cdot \ln t dt$

$u = \ln t$, $du = 1/t$, $dv = t^3 dt$, $v = t^3/3$

Aplicando a fórmula:

$\int u dv = \ln t \cdot t^3/3 + \int t^3/3 \cdot 1/t dt \rightarrow \int u dv = t^3/3 \cdot \ln t + \int t^2/3 dt$

$\int u dv = t^3/3 \cdot \ln t + \frac{1}{3} \int t^2 dt \rightarrow \int u dv = t^3/3 \cdot \ln t + \frac{1}{3} \cdot \frac{t^3}{3}$

$\int u dv = t^3/3 \cdot \ln t + \frac{t^3}{9} + C$

d) $\int \theta \cos \theta d\theta$, $u = \theta$, $dv = \cos \theta d\theta$

$u = \theta$, $du = 1 d\theta$, $dv = \cos \theta d\theta \rightarrow v = \operatorname{sen} \theta$

Aplicando a fórmula:

$\int u dv = \theta \cdot \operatorname{sen} \theta - \int \operatorname{sen} \theta \cdot 1 d\theta$

$\int u dv = \theta \cdot \operatorname{sen} \theta - \cos \theta + C$

5) Use a integração por partes para resolver as integrais abaixo.

a) $\int x \ln x \, dx$

$u = \ln x, du = 1/x, dv = x \cdot dx, v = x^2/2$

Aplicando a fórmula.

$$\int u \cdot dv = \ln x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx \rightarrow \int u \cdot dv = \frac{x^2}{2} \cdot \ln x - \int \frac{x^2}{2x} dx$$

$$\int u \cdot dv = \frac{x^2}{2} \cdot \ln x - \frac{1}{2} \cdot \frac{x^2}{2} \rightarrow \int u \cdot dv = \frac{x^2}{2} \cdot \ln x - \frac{x^2}{4} + C \rightarrow \text{Primitiva}$$

b) $\int_0^1 x e^x \, dx$

$u = x, du = 1 dx, dv = e^x dx, v = e^x$

Aplicando a fórmula.

$$\int u \cdot dv = x \cdot e^x - \int e^x \cdot dx \rightarrow \int u \cdot dv = x \cdot e^x - e^x + C \rightarrow \text{Primitiva}$$

Aplicando intervalo

$$((1 \cdot e^1 - e^1) - (0 \cdot e^0 - e^0)) \rightarrow ((0) - (-1)) \rightarrow 1$$

c) $\int x \cdot \cos 5x \, dx$

$u = x, du = dx, dv = \cos 5x \cdot dx, v = \frac{1}{5} \cdot \sin(5x)$

Aplicando a fórmula.

$$\int u \cdot dv = x \cdot \frac{1}{5} \cdot \sin(5x) - \int \frac{1}{5} \cdot \sin(5x) \, dx$$

$$\int u \cdot dv = \frac{x}{5} \cdot \sin(5x) - \frac{1}{5} \cdot \int \sin(5x) \, dx \rightarrow \int u \cdot dv = \frac{x}{5} \cdot \sin(5x) - \frac{1}{5} \cdot \left(-\frac{1}{5} \cdot \cos u\right)$$

$$\int u \cdot dv = \frac{x}{5} \cdot \sin(5x) - \left(\frac{1}{5} \cdot -\frac{1}{5} \cdot \cos u\right) \rightarrow \left\{ \frac{x}{5} \cdot \sin(5x) + \frac{1}{25} \cdot \cos 5x + C \right\}$$

$$d) \int (x^2 + 2x) \cdot \cos x \, dx$$

$$u = (x^2 + 2x), \, du = (2x + 2), \, dv = \cos x \, dx, \, v = \sin x$$

Aplicando fórmula

$$\int u \cdot dv = (x^2 + 2x) \cdot \sin x - \int \sin x \cdot (2x + 2) \, dx$$

$$\int u \cdot dv = (x^2 + 2x) \cdot \sin x - \text{Aplicar outra vez por partes.}$$

$$\rightarrow u = (2x + 2), \, du = 2 \, dx, \, dv = \sin x \, dx, \, v = -\cos x$$

$$-(2x + 2) \cdot \cos x - \int \cos x \cdot 2 \, dx$$

$$-(2x + 2) \cdot \cos x - (2 \cdot \sin x)$$

$$-(2x + 2) \cdot \cos x + 2 \sin x$$

$$\int u \cdot dv = (x^2 + 2x) \cdot \sin x + (2x + 2) \cdot \cos x - 2 \sin x$$

$$\text{Primitiva} = (x^2 + 2x) \cdot \sin x + (2x + 2) \cdot \cos x - 2 \sin x + C$$

$$e) \int t \sec^2 2t \, dt$$

$$u = t, \, du = 1 \, dt, \, dv = \sec^2 2t \, dt, \, v = \frac{1}{2} \cdot \tan(2t)$$

Aplicando a fórmula

$$\int u \cdot dv = t \cdot \frac{1}{2} \cdot \tan(2t) - \int \frac{1}{2} \cdot \tan(2t) \cdot 1 \, dt$$

$$\int u \cdot dv = \frac{t}{2} \cdot \tan(2t) - \left(\frac{1}{2} \cdot \int \tan(2t) \, dt \right)$$

$$\int u \cdot dv = \frac{t}{2} \cdot \tan(2t) - \left(\frac{1}{2} \cdot -\frac{1}{2} \cdot \ln|\cos(2t)| \right)$$

$$\int u \cdot dv = \frac{t}{2} \cdot \tan(2t) + \frac{1}{4} \cdot \ln|\cos(2t)| + C$$

$$f) \int_0^{\pi/2} (x+1) \cdot \cos 2x \, dx$$

$$u = (x+1), du = 1 \, dx, dv = \cos 2x \, dx, v = \frac{1}{2} \sin(2x)$$

Aplicando a fórmula:

$$\int u \cdot dv = (x+1) \cdot \frac{1}{2} \sin(2x) - \int \frac{1}{2} \sin(2x) \cdot 1 \, dx$$

$$\int u \cdot dv = \frac{(x+1) \cdot \sin(2x)}{2} - \left(\frac{1}{2} \cdot \int \sin(2x) \, dx \right)$$

$$\int u \cdot dv = \frac{(x+1) \cdot \sin(2x)}{2} - \left(\frac{1}{2} \cdot -\frac{1}{2} \cdot \cos(2x) \right)$$

$$\int u \cdot dv = \frac{(x+1) \cdot \sin(2x)}{2} + \frac{1}{4} \cdot \cos(2x) + C \rightarrow \text{Primitiva}$$

Intervalo de $0, \pi/2$.

$$\frac{(\pi/2 + 1) \cdot \sin(\pi) + \frac{1}{4} \cdot \cos(\pi)}{2} - \left(\frac{(0+1) \cdot \sin 0 + \frac{1}{4} \cdot \cos(0)}{2} \right)$$

$$\frac{(\pi/2 + 1) \cdot 0 + \frac{1}{4} \cdot -1}{2} - \left(\frac{1 \cdot 0 + \frac{1}{4} \cdot 1}{2} \right)$$

$$-\frac{1/4}{2} - \frac{1/4}{2} \rightarrow -\frac{2/4}{2} \rightarrow -\frac{1}{2}$$

$$g) \int_0^{\pi} e^x \cdot \cos x \, dx$$

$$u = \cos x, du = -\sin x, dv = e^x \, dx, v = e^x$$

Aplicando fórmula

$$\int u \cdot dv = (\cos x \cdot e^x + \sin x \cdot e^x) dx$$

Aplicando 2º integral por partes

$$u = \sin x, du = \cos x \, dx, dv = e^x \, dx, v = e^x$$

$$\int u \cdot dv = (\sin x \cdot e^x - \int e^x \cdot \cos x \, dx)$$

Volando na primeira

$$\int e^x \cos x = \cos x \cdot e^x + \sin x \cdot e^x - \int e^x \cos x \, dx$$

$$2 \int e^x \cos x = \cos x \cdot e^x + \sin x \cdot e^x \rightarrow \int e^x \cos x = (\cos x \cdot e^x + \sin x \cdot e^x) / 2 + C$$

Intervalo de $0, \pi$

$$\frac{\cos \pi \cdot e^{\pi} + \sin \pi \cdot e^{\pi}}{2} - \frac{\cos 0 \cdot e^0 + \sin 0 \cdot e^0}{2} = \frac{-e^{\pi}}{2} - \frac{1}{2} = -\frac{(e^{\pi} + 1)}{2}$$

$$h) \int_1^3 r^3 \ln r \, dr$$

$$u = \ln r, du = 1/r \, dr, dv = r^3 \, dr, v = r^4/4$$

Aplicando formula

$$\int u \, dv = \ln r \cdot \frac{r^4}{4} - \int \frac{r^4}{4} \cdot \frac{1}{r} \, dr$$

$$\int u \, dv = \ln(r) \cdot \frac{r^4}{4} - \left(\frac{1}{4} \int r^3 \, dr \right) \rightarrow \ln(r) \cdot \frac{r^4}{4} - \left(\frac{1}{4} \cdot \frac{r^4}{4} \right)$$

$$\int u \, dv = \ln(r) \cdot \frac{r^4}{4} - \frac{r^4}{16} + C \rightarrow \frac{r^4}{4} \cdot \ln(r) - \frac{r^4}{16} + C$$

Intervalo de $[1, 3]$

$$\left(\frac{3^4}{4} \cdot \ln(3) - \frac{3^4}{16} \right) - \left(\frac{1^4}{4} \cdot \ln(1) - \frac{1^4}{16} \right)$$

$$\left(\frac{81}{4} \cdot \ln(3) - \frac{81}{16} \right) - \left(\frac{1}{4} \cdot \ln(1) - \frac{1}{16} \right) \rightarrow \frac{81}{4} \cdot \ln(3) - \frac{81}{16} - \frac{1}{4} \cdot 0 + \frac{1}{16}$$

$$\frac{81}{4} \cdot \ln 3 - \frac{80}{16} \rightarrow \left(\frac{81}{4} \cdot \ln 3 - 5 \right)$$

$$1) \int_0^1 (x^2 + 1) e^{-x} \, dx$$

$$u = (x^2 + 1), du = 2x \, dx, dv = e^{-x} \, dx, v = -e^{-x}$$

Aplicando formula

$$\int u \, dv = (x^2 + 1) \cdot (-e^{-x}) - \int (-e^{-x}) \cdot 2x \, dx$$

Integrando por partes 2º

$$u = 2x, du = 2, dv = -e^{-x} \, dx, v = e^{-x}$$

$$\int u \, dv = 2x \cdot e^{-x} - \int e^{-x} \cdot 2 \, dx$$

$$2x \cdot e^{-x} - (2 \cdot (-e^{-x})) \rightarrow (2x \cdot e^{-x} + 2e^{-x})$$

Juntando

$$(x^2 + 1) \cdot (-e^{-x}) - (2x \cdot e^{-x} + 2e^{-x}) \rightarrow -(x^2 + 1) \cdot e^{-x} - 2x \cdot e^{-x} - 2e^{-x}$$

Intervalo $[0, 1]$

$$-(1^2 + 1) \cdot e^{-1} - 2 \cdot 1 \cdot e^{-1} - 2e^{-1} - (-(0^2 + 1) \cdot e^{-0} - 2 \cdot 0 \cdot e^{-0} - 2e^{-0})$$

$$-\frac{2}{e} - \frac{2}{e} - \frac{2}{e} + 3 \rightarrow \left(-\frac{6}{e} + 3 \right)$$

$$J) \int_0^{\pi} \sin^3 x \, dx \rightarrow \sin^2 x \cdot \sin x \, dx$$

$$u = \sin^2 x, \, du = 2 \sin x \cos x \, dx, \, dv = \sin x \, dx, \, v = -\cos(x)$$

Aplicando fórmula

$$S u \cdot dv = \sin^2 x \cdot (-\cos(x)) - \int -\cos x \cdot 2 \sin x \cos x \, dx = -(2 \int \sin^2 x \cos^2 x \, dx)$$

Aplicando método da substituição

$$\int \cos^2 x \sin x \, dx \rightarrow u = \cos x, \, du = -\sin x \, dx$$

$$\int u^2 \cdot (-du) \rightarrow -\frac{u^3}{3} \rightarrow -\frac{\cos^3 x}{3}$$

Então

$$-\sin^2 x \cos x - \left(-2 \cdot -\frac{\cos^3 x}{3} \right) \rightarrow -\sin^2 x \cos x + \frac{2(\cos^3 x)}{3} + C$$

Intervalo de 0 a π

$$-\sin^2 \pi \cos \pi + \frac{2 \cos^3 \pi}{3} - \left(-\sin^2 0 \cos 0 + \frac{2 \cos^3 0}{3} \right)$$

$$-2/3 \cdot -1 - (-2/3 \cdot 1) \rightarrow 2/3 + 2/3 = 4/3$$

$$K) \int \sqrt{x} e^{\sqrt{x}} \, dx \quad u = t$$

$$t = \sqrt{x} \rightarrow dt = \frac{1}{2\sqrt{x}} dx = \frac{1}{2} dt \, dx$$

$$\int t e^t \cdot 2 \, dt = 2 \int t^2 \cdot e^t \, dt$$

$$u = t^2, \, du = 2t \, dx, \, dv = e^t \, dt, \, v = e^t$$

Aplicando fórmula

$$S u \cdot dv = t^2 \cdot e^t - \int e^t \cdot 2t \, dx \rightarrow t^2 \cdot e^t - 2 \int t \cdot e^t \, dx$$

$$\int e^t \cdot t \, dx \rightarrow u = t, \, du = dt, \, dv = e^t \, dx, \, v = e^t$$

$$2t \cdot e^t - 2 \int e^t \cdot t \, dx \rightarrow (2t \cdot e^t - 2e^t)$$

Voltando ...

$$2t^2 \cdot e^t - (2t \cdot e^t - 2e^t) \rightarrow 2t^2 \cdot e^t - 2t \cdot e^t + 2e^t \rightarrow 2e^t (t^2 - t + 1)$$

Voltando

$$2e^{\sqrt{x}} (\sqrt{x}^2 - 2\sqrt{x} + 2) \rightarrow 2e^{\sqrt{x}} (x - 2\sqrt{x} + 2)$$

⑥ Uma força de amortecimento, causada pelo amortecedor representado na figura abaixo, descreve o movimento oscilatório de uma massa acoplada a uma mola sob a ação da gravidade.

Sabendo que a posição da massa no tempo t é

$$y = 2e^{-t} \cdot \sin t$$

para $t \geq 0$, onde $y=0$ é a posição de equilíbrio, encontre o valor médio de y sobre o intervalo $0 \leq t \leq 2\pi$

$$u = \sin x, du = \cos x dx, dv = 2e^{-x} dx, v = -2e^{-x}$$

Aplicando fórmula

$$\int u \cdot dv = u \cdot v - \int v \cdot du$$

$$= -2e^{-x} \cdot \sin x + 2 \int e^{-x} \cdot \cos x dx$$

2º Integração por partes

$$\int 2e^{-x} \cdot \cos x dx = u = \cos x, du = -\sin x dx, dv = 2e^{-x} dx, v = -2e^{-x}$$

Aplicando fórmula

$$\int u \cdot dv = u \cdot v - \int v \cdot du$$

Voltando

$$\int 2e^{-x} \cdot \sin x dx = -2e^{-x} \cdot \sin x + \cos x \cdot 2e^{-x} - 2 \int e^{-x} \sin x dx$$

$$I = -2e^{-x} \cdot \sin x + \cos x \cdot 2e^{-x} - I$$

$$2I = -2e^{-x} \cdot \sin x + \cos x \cdot 2e^{-x}$$

$$I = -e^{-x} \cdot \sin x + e^{-x} \cdot \cos x$$

Intervalo $(0, 2\pi)$

$$(e^{-2\pi} \cdot \sin 2\pi + e^{-2\pi} \cdot \cos 2\pi) - (e^{-0} \cdot \sin 0 + e^{-0} \cdot \cos 0)$$

$$= -e^{-2\pi} - 1$$

Fórmula do Médio

$$\frac{1}{2\pi} \cdot (-e^{-2\pi} - 1) \rightarrow$$

$$\frac{-e^{-2\pi} - 1}{2\pi}$$